

Graph Neural Networks

Introduction

A Graph Neural Network (GNN) is a neural network which simply takes graphs as input. The key idea here is that since the input data has graph structure, the model can be designed to exploit inductive biases. Many datasets such as social networks, traffic flow, and physics simulations can all take advantage of this inductive bias.

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be the usual definition of a graph where, for simplicity, we assume that the graph is undirected. Here, define $N = |\mathcal{V}|$. Let $\mathcal{A}$ be the symmetric adjacency matrix of $\mathcal{G}$. As with many deep learning models, we will have input embeddings of the nodes. For node i , let $\mathbf{x}_i$ denote its D -dimensional embedding vector. We can stack these embedding vectors into a $N \times D$ data matrix $\mathbf{X}$ such that:

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_N^T \end{pmatrix}$$

One property we want our model to have is *permutation invariance*. This means that the output of the neural network should not depend on the labeling of the nodes. In other words, let $\mathbf{P}$ be a $N \times N$ permutation matrix. If we permute the data $\tilde{\mathbf{X}} = \mathbf{P}\mathbf{X}$ and permute the labels of the nodes as

$$\tilde{\mathcal{A}} = \mathbf{P}\mathcal{A}\mathbf{P}^T$$

then, defining our neural network to be y_θ with learnable parameters θ , the following must hold:

$$y_\theta(\tilde{\mathbf{X}}, \tilde{\mathcal{A}}) = y_\theta(\mathbf{X}, \mathcal{A})$$

This is called the *invariance property*. It tells us, under any arbitrary permutation of the labels on the nodes, the neural network must give the same output.

Furthermore, we may want to make predictions associated to each node. A good example of this is if we are given a large social network graph as input and we must decide if each person is either Democrat or Republican. This is a binary classification task on each node in the input graph. In that case, the predictions should not be dependent on the labeling of the nodes. To this end, we can formulate the desired property as follows:

Let $\mathbf{y}(\cdot, \cdot)$ define the $N \times 1$ vector of predictions where each element is associated with a node in the graph. We wish that:

$$\mathbf{y}(\tilde{\mathbf{X}}, \tilde{\mathcal{A}}) = \mathbf{P}\mathbf{y}(\mathbf{X}, \mathcal{A})$$

This is called the *equivariance property*